

FIXLINE

P5-B MVP PROGRESS REPORT

Campus Helpdesk &
Maintenance Tickets

Current delivery snapshot

Foundation is in place.

The repository, Dockerized PostgreSQL environment, and first CI pipeline are working. Backend, mobile, data, AI, and UI/UX workstreams are progressing in

REPORT SCOPE

What each P5-B member has completed or is currently implementing for the MVP.

MVP DELIVERY SNAPSHOT

20 SEP 2026

TEAM PROGRESS

One MVP, six parallel workstreams

Current responsibilities are aligned around a single campus helpdesk flow: access, data, interface, AI suggestions, and deployable infrastructure.

Abdelrahman Hatem

DevOps

Foundation ready

DONE

Hazem Mahmoud

Backend

Core access layer

IN PROGRESS

Ahmed Wael Nashat

Mobile

Login + dashboard

IN PROGRESS

Radwa Mohamed

Data Analysis

ERD + KPI data

IN PROGRESS

Waseem Medhat Ramzy

AI Engineering

FastAPI rule service

IN PROGRESS

Yara Salah

UI/UX

Sign-in screen

DONE

CURRENT INTEGRATION DEPENDENCIES

- Hazem + Radwa: align ERD/schema requirements before the operational database structure is finalized.
- Ahmed -> Hazem: login and dashboard depend on authentication endpoints and usable backend data.
- Waseem -> Hazem: FastAPI rule-based suggestions integrate through the backend contract.
- Abdelrahman -> Hazem: backend containerization and CI build/test steps start when runtime commands, ports, migrations, seed and health endpoint are available.
- Yara -> Product implementation: UI/UX screens define the interaction direction and implementation handoff.

DEVOPS

Abdelrahman Hatem

FOUNDATION READY

The DevOps foundation is operational: the repository is organized, PostgreSQL runs reproducibly in Docker, and the first CI workflow is green on GitHub Actions.

WORK COMPLETED / CURRENT OUTPUT

- Created the GitHub repository and standardized the project structure.
- Established the working branch flow and cleaned up the repository structure.
- Created docker-compose.yml and launched PostgreSQL 16 in Docker.
- Added PostgreSQL health checks and verified readiness as healthy.
- Verified database access using psql against campus_helpdesk.
- Configured and verified a persistent PostgreSQL volume.
- Protected the local .env from Git and prepared .env.example.
- Built the initial GitHub Actions CI workflow.
- CI now validates Docker Compose, starts PostgreSQL, waits for readiness, tests DB connectivity, and cleans up.
- Verified the CI workflow successfully on GitHub Actions.

NEXT INTEGRATION STEP

Document the DevOps workflow now; integrate the backend container and backend CI steps after Hazem provides runtime commands, port, migrations, seed and health endpoint.

BACKEND

Hazem Mahmoud

IN PROGRESS

Hazem is building the operational backend foundation that will enforce data structure, user authentication, and role-based access for the site.

WORK COMPLETED / CURRENT OUTPUT

- Working on the backend schema structure.
- Setting up and implementing the project database.
- Implementing authentication for site access.
- Implementing authorization and role access logic.
- Translating the agreed data/ERD design into the operational backend data layer.

NEXT INTEGRATION STEP

Provide stable authentication endpoints and runtime details so Mobile and DevOps can integrate without guessing backend behavior.

MOBILE APP DEVELOPMENT

Ahmed Wael Nashat

IN PROGRESS

Ahmed is implementing the first user-facing mobile flow, focusing on access and the first post-login dashboard experience.

WORK COMPLETED / CURRENT OUTPUT

- Implementing the Login flow.
- Implementing the Dashboard.
- Preparing the app flow to consume backend authentication and dashboard data once endpoints are ready.

NEXT INTEGRATION STEP

Connect the login and dashboard screens to Hazem's authentication and dashboard APIs when the backend contract is ready.

DATA ANALYSIS

Radwa Mohamed

IN PROGRESS

Radwa is defining the data foundation and preparing the analytical layer that will support database alignment, demo data, KPIs, and project insights.

WORK COMPLETED / CURRENT OUTPUT

- Building the project ERD.
- Supporting schema definition and field alignment.
- Preparing fake / synthetic data for safe MVP testing.
- Performing data cleaning preparation.
- Defining the project KPIs.
- Preparing insights that can feed dashboards and reporting.

NEXT INTEGRATION STEP

Align the ERD/schema with Hazem and ensure KPI fields exist in the operational database before dashboard metrics are finalized.

AI ENGINEERING

Waseem Medhat Ramzy

IN PROGRESS

Waseem is building the MVP AI layer as a lightweight FastAPI service using deterministic rule-based functions rather than a trained model.

WORK COMPLETED / CURRENT OUTPUT

- Setting up the FastAPI AI service.
- Defining the role and responsibilities of the AI service inside the MVP.
- Building rule-based service functions for deterministic suggestions.
- Preparing the service contract needed for backend integration.
- Keeping the MVP AI logic explainable and suitable for fallback behavior.

NEXT INTEGRATION STEP

Expose a stable FastAPI contract for category/priority suggestions so Hazem can integrate it into the backend workflow.

UI/UX - FINAL TEAM SECTION

Yara Salah

SIGN-IN SCREEN READY

Yara designed the first authentication screen for the Fixline campus helpdesk interface, establishing the visual direction for the product.

DESIGN SUMMARY

- Split-screen layout: dark navy branding panel on the left and a clean sign-in form on the right.
- Brand: Fixline, with the message “Report it. Track it. See it fixed.”
- Authentication fields: email and password, plus a Keep me signed in option.
- Actions: primary Sign in button, Forgot password link, and Continue with University option.
- Visual direction: simple, modern and professional, appropriate for a campus helpdesk interface.
- Integration note: University SSO and password-reset behavior depend on the backend MVP authentication decisions.